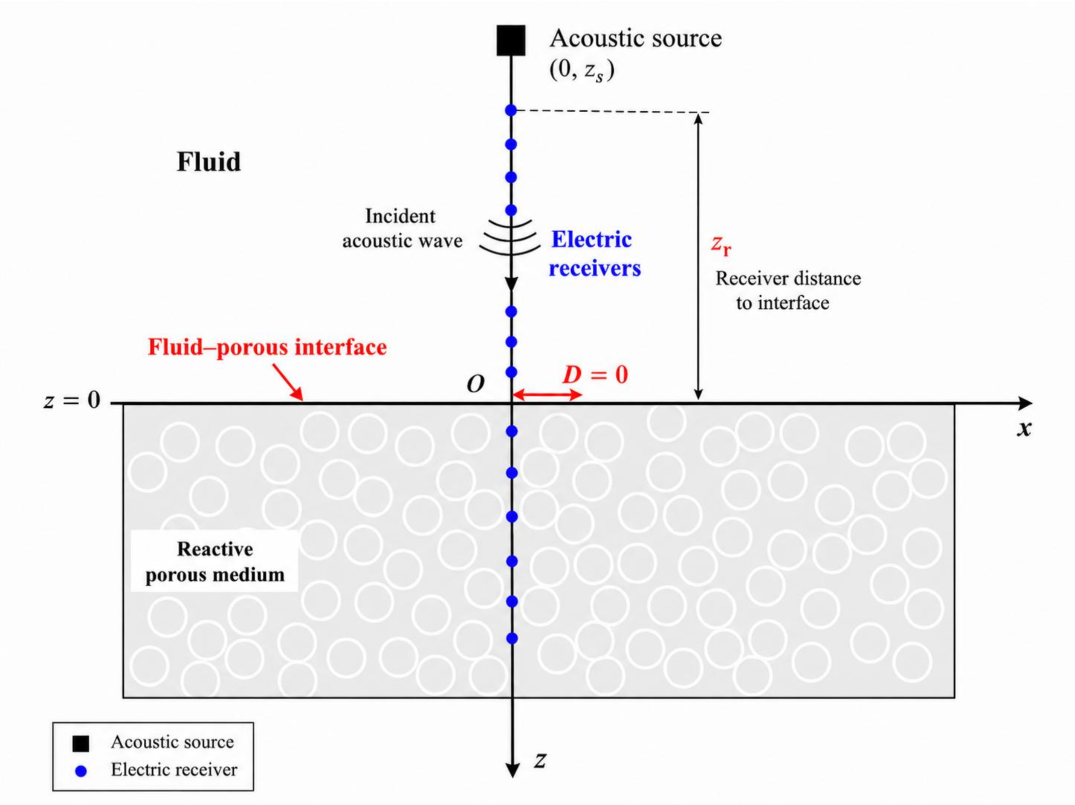
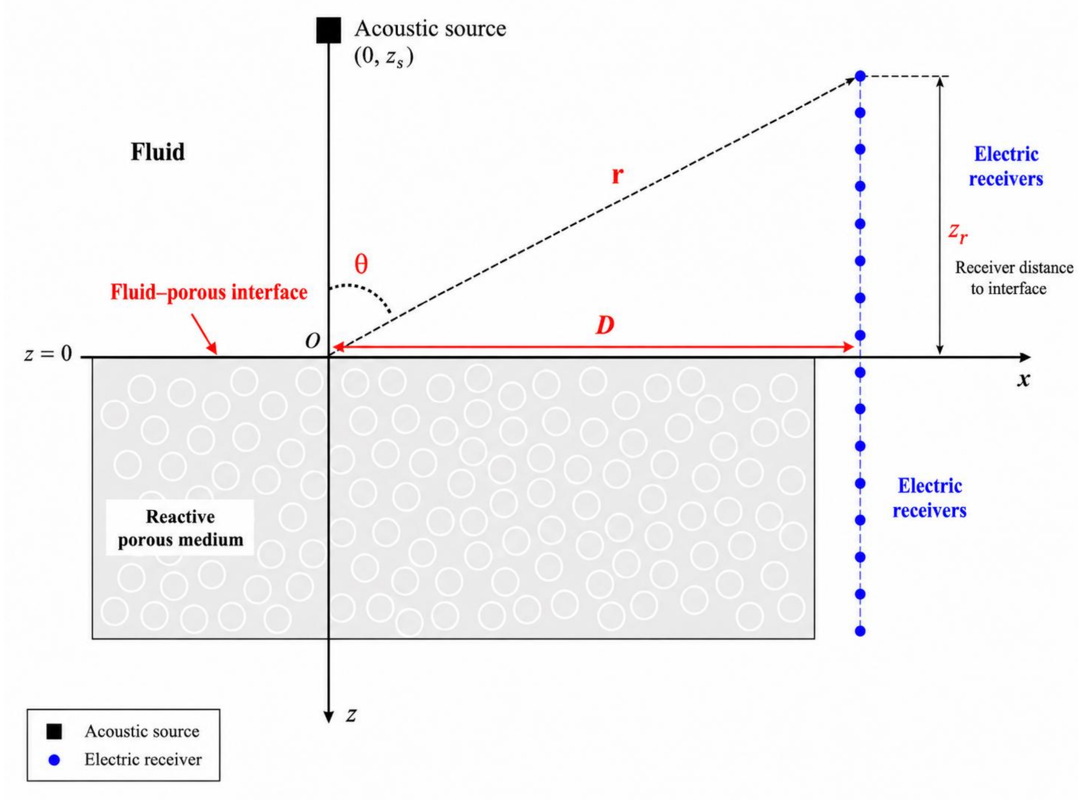
RTM结合整电模拟

几何模型

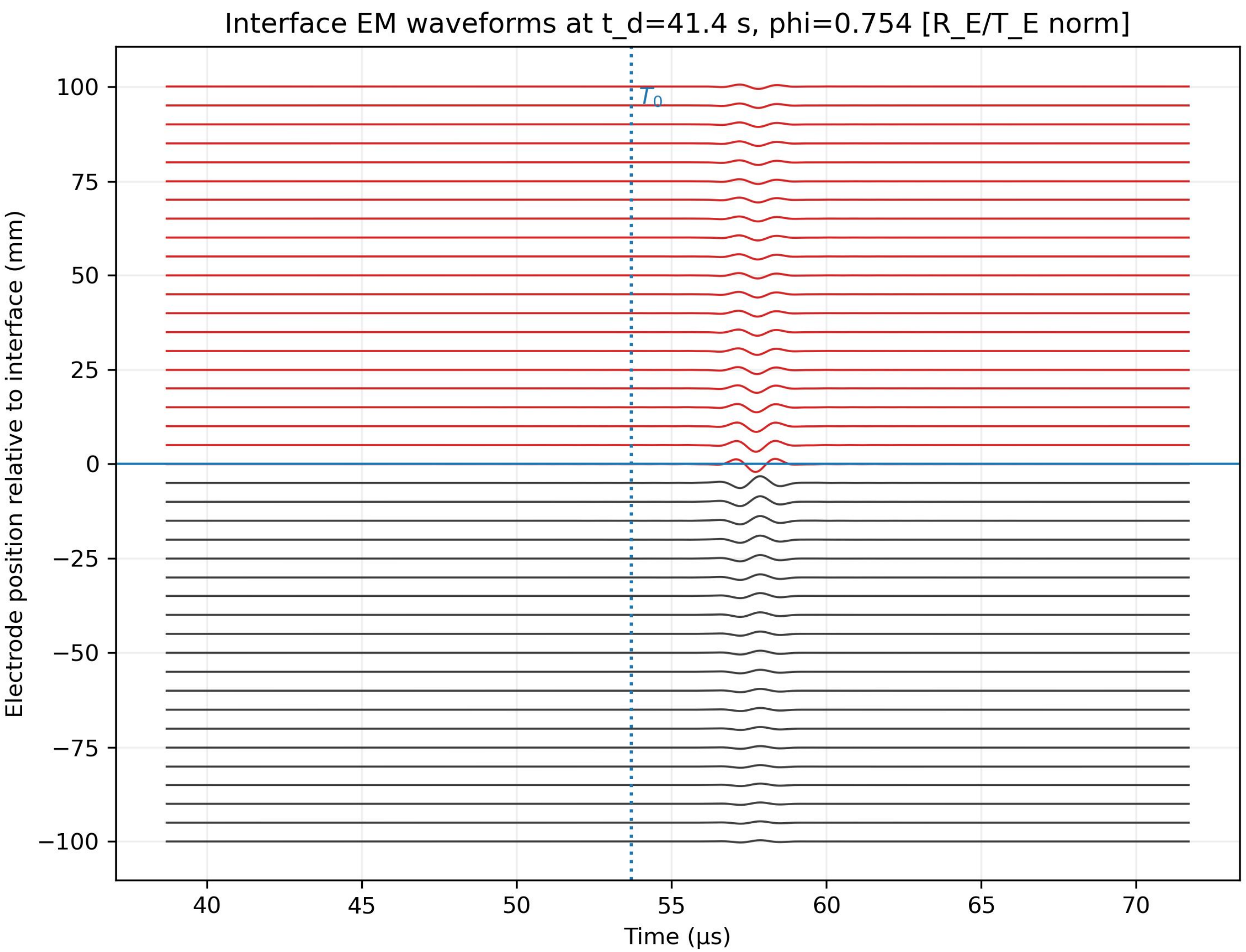


Acoustic source
(0, z_s)
Fluid
Incident
acoustic wave
Electric
receivers
z_r
Receiver distance
to interface
Fluid–porous interface
O
$D = 0$
$z = 0$
x
Reactive
porous medium
Acoustic source
Electric receiver
z



Acoustic source
(0, z_s)
Fluid
r
Electric
receivers
z_r
Receiver distance
to interface
θ
Fluid–porous interface
O
D
$z = 0$
x
Reactive
porous medium
Electric
receivers
z
Acoustic source
Electric receiver

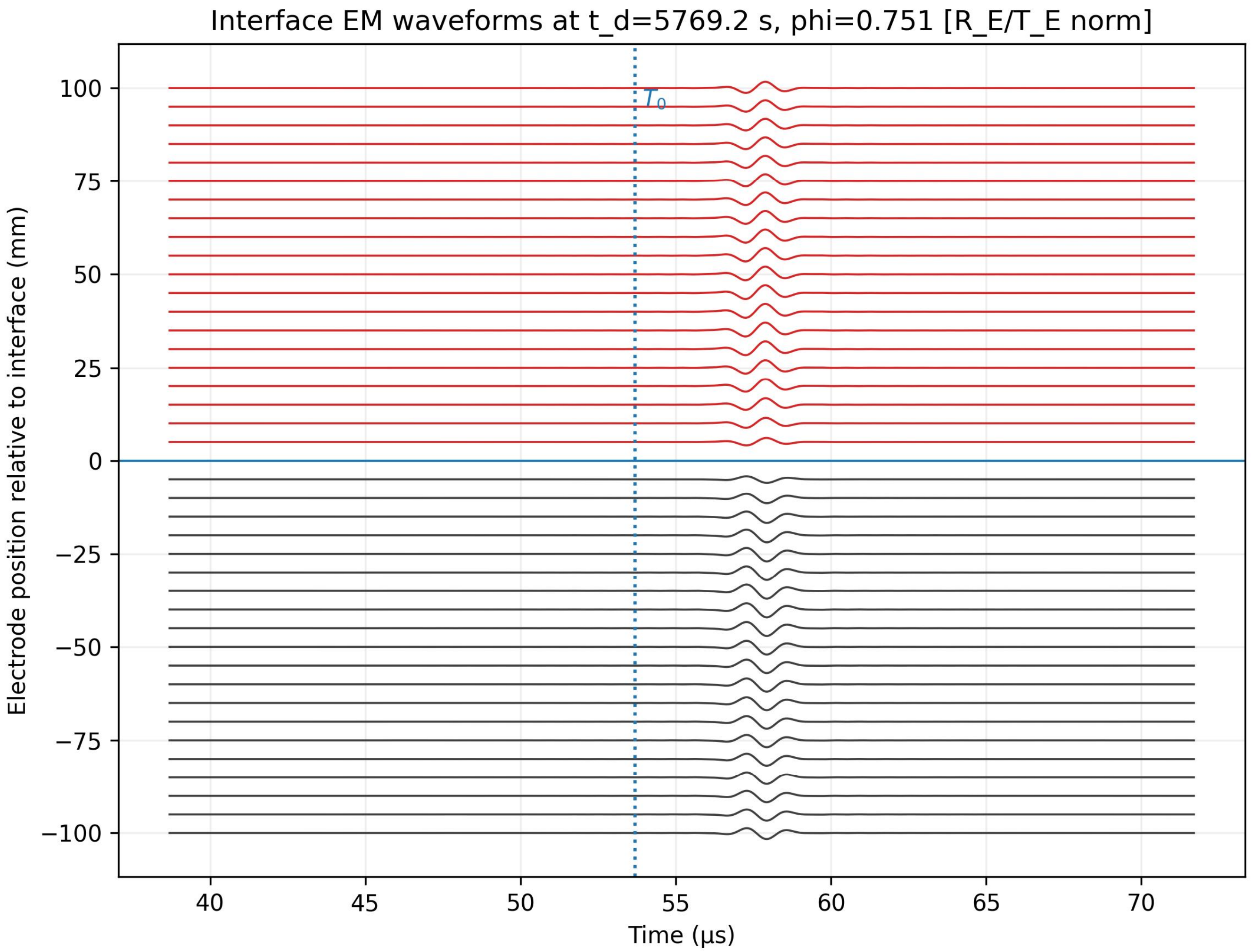
零偏移距震电波形结果



构建了和(Liu et al., 2018)相同的震电理论模型

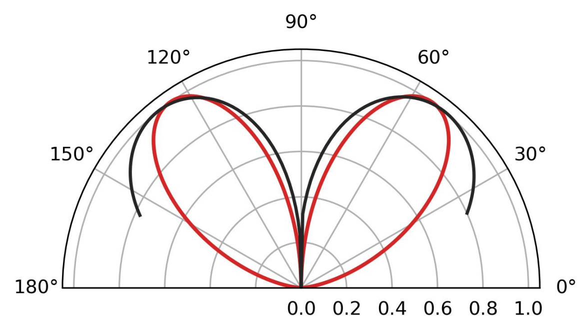
- 所有 interface EM 信号同时到达，说明它们来自界面转换
- 幅值越靠近界面越大
- 界面上下信号发生极性反转

有限偏移距震电波形结果

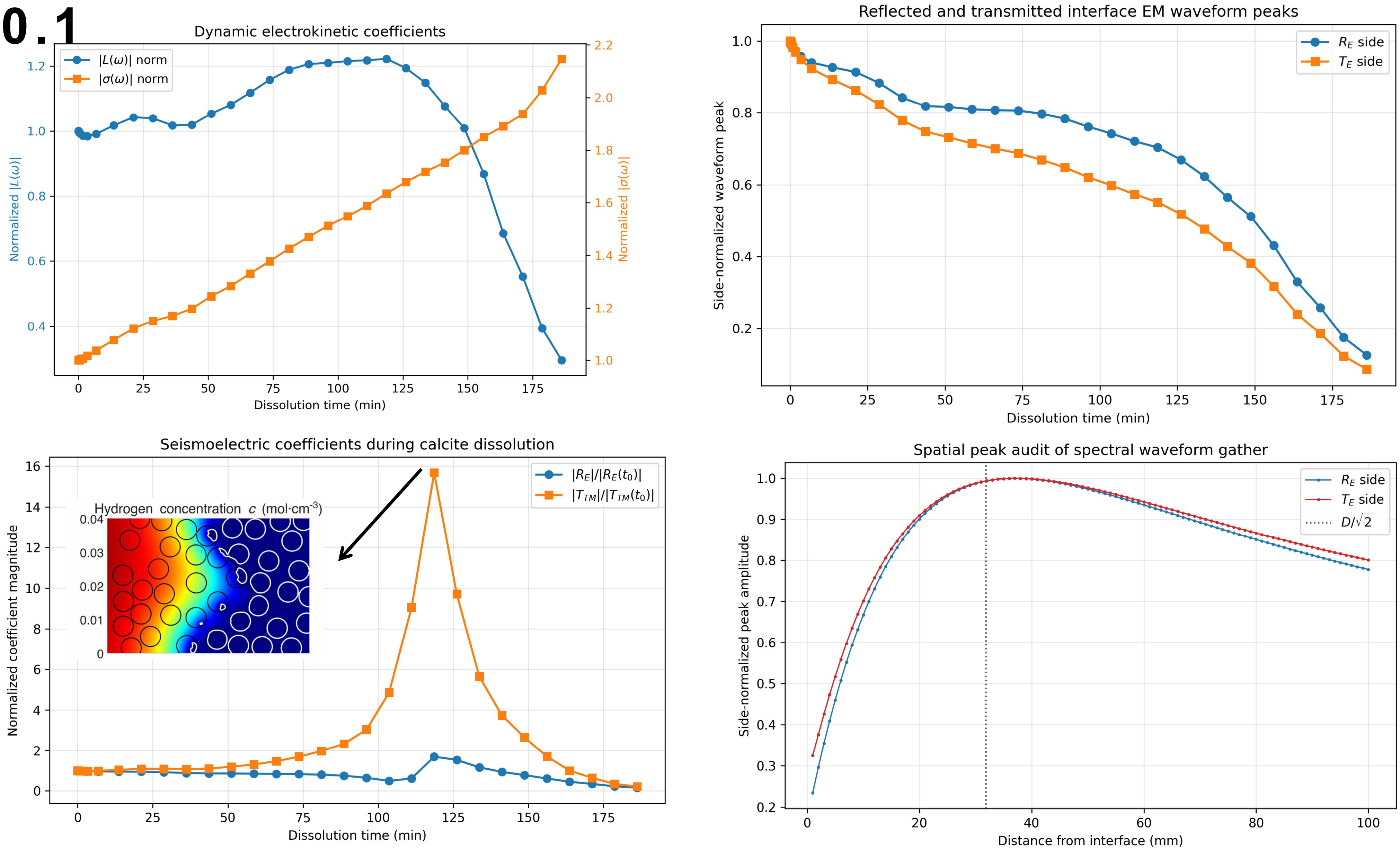


构建了和(Liu et al., 2018)相同的震电理论模型

- 所有 interface EM 信号同时到达，说明它们来自界面转换
- 界面上下信号发生极性反转，并且幅值跨界面连续变化(电偶极子模型)
- 有限偏移距下幅值不是越靠近界面越大，而是在两侧呈非单调分布

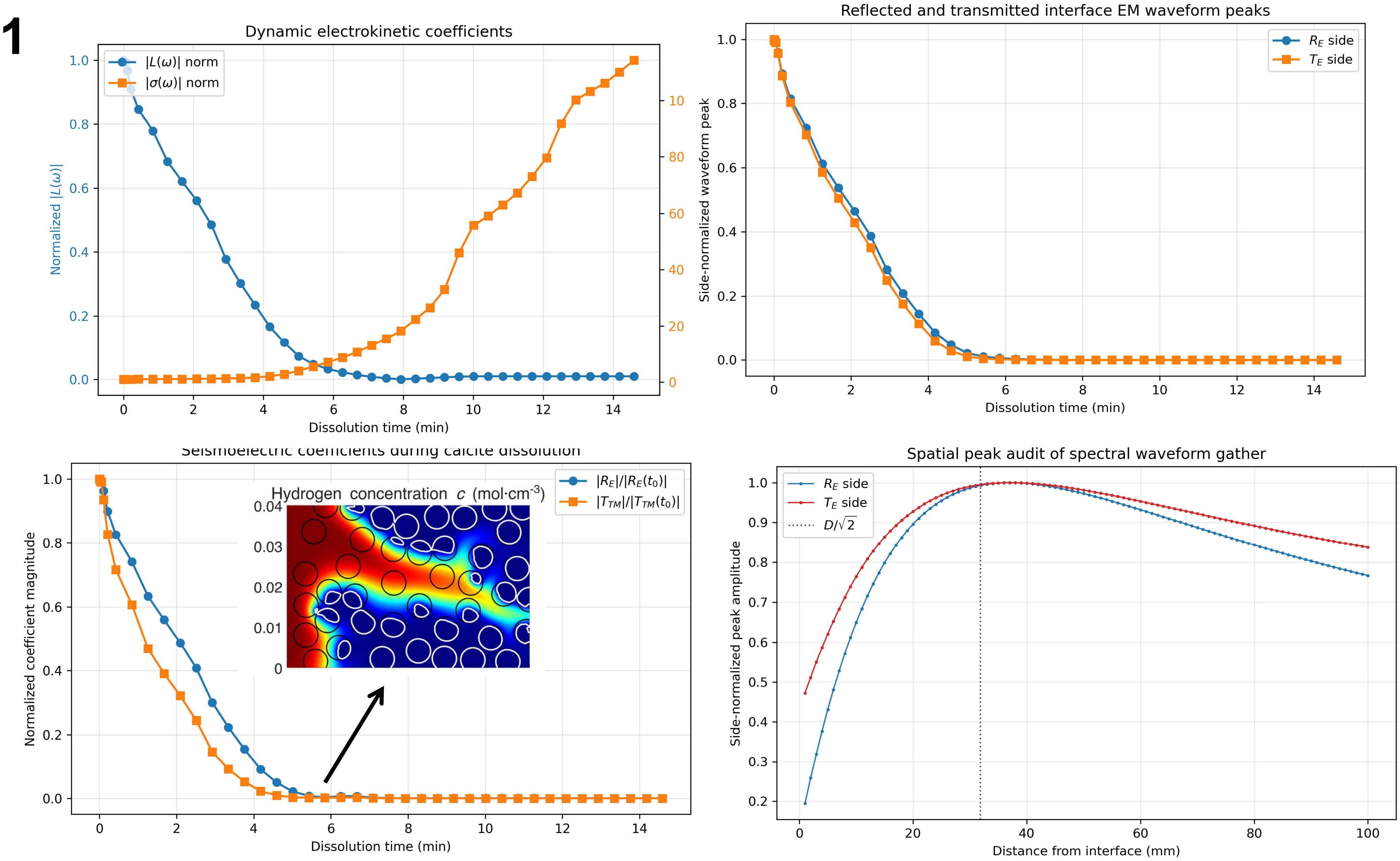


Pe0.1



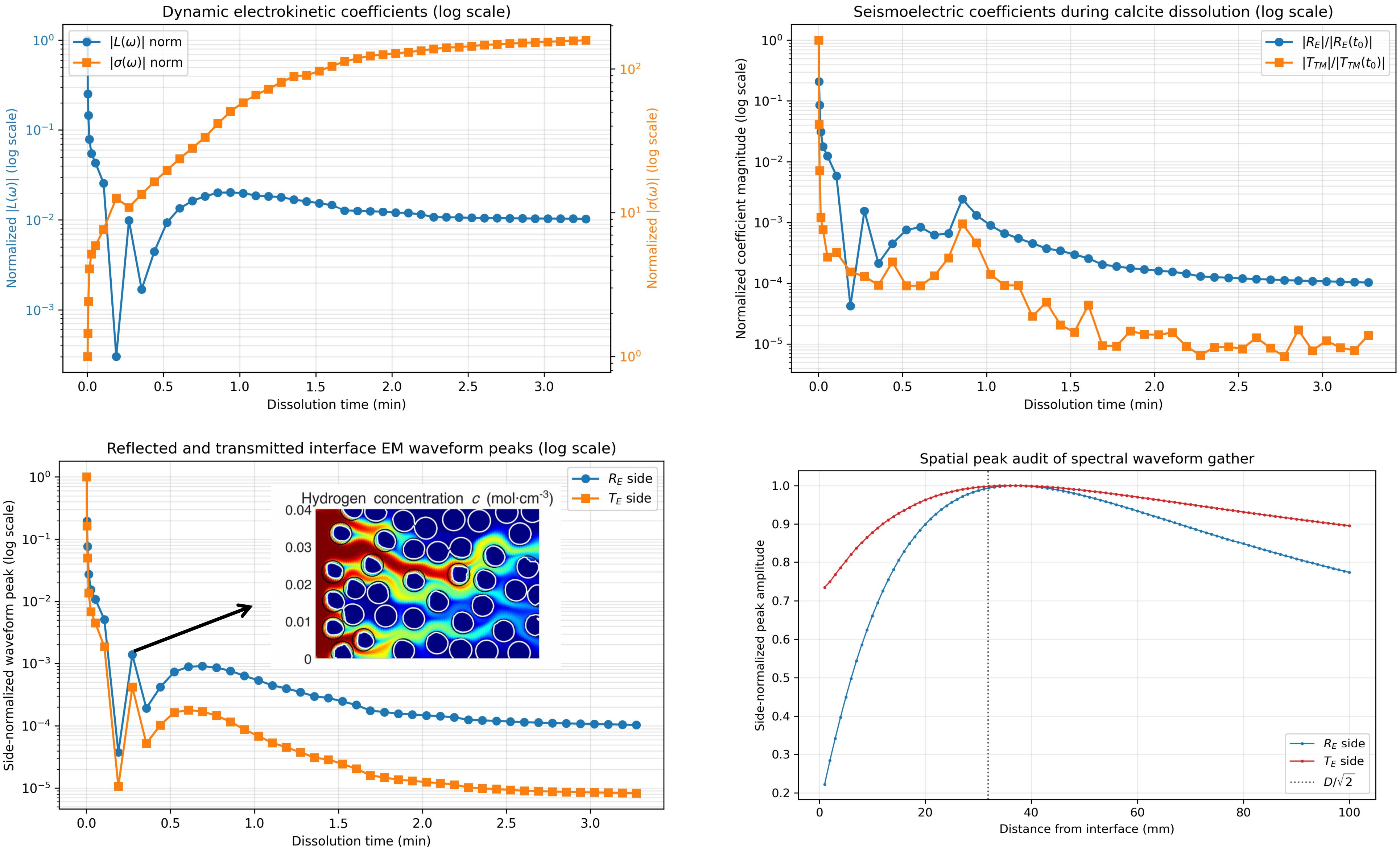
Dynamic electrokinetic coefficients
|L(ω)| norm
|σ(ω)| norm
Normalized |L(ω)|
Normalized |σ(ω)|
Dissolution time (min)
Reflected and transmitted interface EM waveform peaks
R_E side
T_E side
Side-normalized waveform peak
Dissolution time (min)
Seismoelectric coefficients during calcite dissolution
$|R_E|/|R_E(t_0)|$
$|T_{TM}|/|T_{TM}(t_0)|$
Hydrogen concentration c (mol·cm^{-3})
Normalized coefficient magnitude
Dissolution time (min)
Spatial peak audit of spectral waveform gather
R_E side
T_E side
$D/\sqrt{2}$
Side-normalized peak amplitude
Distance from interface (mm)

Pe1



Dynamic electrokinetic coefficients
|L(ω)| norm
|σ(ω)| norm
Normalized |L(ω)|
Dissolution time (min)
Reflected and transmitted interface EM waveform peaks
R_E side
T_E side
Side-normalized waveform peak
Dissolution time (min)
Seismoelectric coefficients during calcite dissolution
$|R_E|/|R_E(t_0)|$
$|T_{TM}|/|T_{TM}(t_0)|$
Hydrogen concentration c (mol·cm^{-3})
Normalized coefficient magnitude
Dissolution time (min)
Spatial peak audit of spectral waveform gather
R_E side
T_E side
$D/\sqrt{2}$
Side-normalized peak amplitude
Distance from interface (mm)

Pe10



Dynamic electrokinetic coefficients (log scale)
|L(ω)| norm
|σ(ω)| norm
Normalized |L(ω)| (log scale)
Normalized |σ(ω)| (log scale)
Dissolution time (min)
Seismoelectric coefficients during calcite dissolution (log scale)
|R_E|/|R_E(t_0)|
|T_TM|/|T_TM(t_0)|
Normalized coefficient magnitude (log scale)
Dissolution time (min)
Reflected and transmitted interface EM waveform peaks (log scale)
R_E side
T_E side
Side-normalized waveform peak (log scale)
Hydrogen concentration c (mol·cm^-3)
0.04
0.03
0.02
0.01
0
Dissolution time (min)
Spatial peak audit of spectral waveform gather
R_E side
T_E side
D/√2
Side-normalized peak amplitude
Distance from interface (mm)